

What is Docker?

Simple Definition

- Docker is an **open-source platform** used to **create, deploy, and run applications inside containers**.
- Docker packages an application with all its dependencies into a **container**, so it runs the same everywhere.

Easy Definition:

Docker is a tool that creates and runs applications inside lightweight containers.

Key Points about Docker

1. Open-Source Platform

- Docker is free to use.
 - Anyone can download, use, and improve it.
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2. Used to Create, Deploy, and Run Applications

- **Create** → Build an application into a container.
 - **Deploy** → Send the container to a server or cloud.
 - **Run** → Start the application from the container.
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3. Uses Containers

- Docker runs applications inside **containers**.
 - A container includes:
 - Application
 - Required libraries
 - Dependencies
 - Configuration files
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4. Uses the Host Operating System

- Docker containers **share the Host OS kernel**.
- Containers do **not** need their own operating system.
- This makes them:
 - Lightweight
 - Fast
 - Efficient

Example:

Host OS (Linux)

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Docker Engine

└─ Container 1

└─ Container 2

└─ Container 3

All containers use the **same Linux kernel**.

5. Docker Can Be Installed on Many Operating Systems

Docker can be installed on:

- Windows
- Linux
- macOS

However:

- Docker Engine runs **natively (directly)** on Linux.
- On Windows and macOS, Docker uses a lightweight virtual machine behind the scenes.

6. Docker is Written in Go Language

- Docker was developed using the **Go (Golang)** programming language.
 - Go is fast, efficient, and widely used for cloud-native tools.
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7. Docker Uses OS-Level Virtualization

- Docker performs **Operating System (OS) level virtualization**, also called **Containerization**.
- Containers share the host OS instead of having a separate operating system.

Difference:

- **Virtual Machine (VM):** Each VM has its own operating system.
 - **Docker Container:** Containers share the host operating system.
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8. Docker Release

- Docker was first released in **March 2013**.
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9. Docker Founders

Docker was created by:

- **Solomon Hykes**
 - **Sébastien Pahl** (often written as Sebastien Pahl in some notes)
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10. Docker vs VMware

Docker

Uses OS-level virtualization

Lightweight

Starts in seconds

Shares Host OS

VMware

Uses Hardware-level virtualization

Heavy

Takes minutes to start

Each VM has its own OS

Docker

VMware

Uses less RAM and storage Uses more RAM and storage

Important Terms

Term	Easy Meaning
Docker	Tool that creates and runs containers
Container	A lightweight package containing an application and its dependencies
Host OS	The operating system on which Docker runs
Kernel	The core part of the operating system shared by containers
Containerization	Running applications inside containers using OS-level virtualization
Open Source	Software whose source code is freely available

Interview/Exam Definition ★

Docker is an open-source platform that uses containerization (OS-level virtualization) to build, deploy, and run applications inside lightweight containers.

Easy Memory Trick

Application



Docker



Container



Runs on Any System

Remember:

-  **Docker = Creates Containers**
-  **Container = Holds the Application**
-  **Kubernetes = Manages Containers**

Great! I'll explain these **Advantages and Disadvantages of Docker** in **easy beginner-friendly bullet points** that are perfect for your **KCNA notes**.

Advantages of Docker

1. No Pre-allocation of RAM

Easy Meaning

- Docker **does not reserve RAM in advance**.
- It uses memory **only when a container needs it**.
- This saves system resources.

Example

Suppose your computer has **8 GB RAM**.

- A Virtual Machine may reserve **2 GB RAM** even if it is idle.
- A Docker container uses RAM **only while running**.

Benefit: Better memory utilization.

2. CI Efficiency (Continuous Integration)

Easy Meaning

- Docker creates an **image** once.
- The same image is used during:
 - Testing
 - Development
 - Staging
 - Production

This ensures the application behaves the same everywhere.

Example

Developer builds an image:

Docker Image



Testing



QA



Production

Same image → Same application → Fewer errors.

3. Less Cost

Easy Meaning

- Docker uses fewer system resources.
- One server can run many containers.
- Less hardware is needed.

Result: Lower infrastructure cost.

4. Lightweight

Easy Meaning

- Containers do **not include a complete operating system**.
- They share the Host OS.

Because of this:

- Smaller size
- Faster startup

- Less disk space
 - Less RAM usage
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5. Can Run on Physical Hardware or Cloud

Easy Meaning

Docker containers can run on:

- Physical Server
- Virtual Machine
- AWS
- Azure
- Google Cloud
- Local Laptop

You don't need to change your application.

6. Image Can Be Reused

Easy Meaning

- Once you create a Docker image,
- You can use it again and again.
- No need to rebuild it every time.

Example

Docker Image



Laptop



AWS



Azure



Google Cloud

The same image works everywhere.

7. Very Fast Deployment

Easy Meaning

Containers start in **seconds**.

Unlike Virtual Machines:

- VM → Takes minutes
- Docker Container → Takes seconds

This helps applications start quickly.

Quick Revision of Advantages

Advantage	Easy Meaning
No Pre-allocation of RAM	Uses memory only when needed
CI Efficiency	Same image is used everywhere
Less Cost	Saves hardware and money
Lightweight	Small and fast
Runs Anywhere	Physical server, VM, or Cloud
Reusable Images	One image can be used many times
Fast Deployment	Containers start in seconds

Disadvantages of Docker

1. Not Good for Rich GUI Applications

Easy Meaning

Docker is mainly designed for:

- Web applications
- APIs
- Backend services
- Databases

It is **not ideal** for applications with heavy graphical interfaces (GUI).

Examples of Rich GUI Applications

- Adobe Photoshop
- AutoCAD
- Video editing software
- 3D games

These applications usually work better directly on the operating system or in a virtual machine.

2. Difficult to Manage a Large Number of Containers

Easy Meaning

Managing a few containers is easy.

Managing hundreds or thousands manually is difficult.

Example

- 5 Containers → Easy to manage
- 500 Containers → Very difficult manually

This is why **Kubernetes** was created.

Remember:

- Docker creates containers.
 - Kubernetes manages large numbers of containers.
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3. Limited Cross-Platform Support

Easy Meaning

Containers share the **host operating system kernel**.

Because of this:

- A **Linux container** cannot run directly on a **Windows kernel**.
- A **Windows container** cannot run directly on a **Linux kernel**.

Docker Desktop can help by using virtualization behind the scenes, but the container itself still depends on a compatible OS kernel.

4. Different Linux Distributions

The slide says:

"Even can't run on different OS (Ubuntu & CentOS)"

This statement is **not completely correct**.

Correct Explanation

- Ubuntu and CentOS are **both Linux distributions**.
- Most Linux containers **can run on different Linux distributions** because they share the Linux kernel.
- Problems may occur only if an application depends on distribution-specific libraries or kernel features.

So for your exams, remember:

Docker containers need a **compatible operating system kernel**, not necessarily the exact same Linux distribution.

Quick Revision of Disadvantages

Disadvantage

Not suitable for Rich GUI

Easy Meaning

Best for backend and server applications

Disadvantage

Easy Meaning

Hard to manage many containers Kubernetes solves this problem

Limited by OS kernel

Containers need a compatible operating system kernel

Distribution-specific issues

Some applications may require specific Linux libraries

★ KCNA Interview Tip

One of the most common questions is:

Q: If Docker is so powerful, why do we need Kubernetes?

Answer:

- Docker is excellent at **creating and running containers**.
- When you have hundreds or thousands of containers, managing them manually becomes difficult.
- **Kubernetes** automates container management by handling scaling, load balancing, self-healing, and deployments. This is why Kubernetes is used together with Docker in modern cloud-native environments.